

Can We Empirically Trace How Burnout Turns into Emotional Debt and Emotional Smells?

Giammaria Giordano

*Dipartimento di Scienze e Tecnologie dell'Informazione
Pegaso University
Naples, Italy
giammaria.giordano@unipegaso.it*

Francesco Maria Torino

*Computer Science
University of Salerno
Fisciano, Italy
f.torino1@studenti.unisa.it*

Fabio Palomba

*Computer Science
University of Salerno
Fisciano, Italy
fpalomba@unisa.it*

Abstract—Burnout is an increasingly prevalent challenge in software engineering, yet its impact on software development processes remains insufficiently understood. In this paper, we introduce Emotional Debt, a novel socio-technical construct that captures the accumulation of unresolved psychological strain experienced by developers. We further operationalize this construct through Emotional Smells, a set of observable behavioral indicators that can manifest within software repositories. To investigate the relationships among burnout, Emotional Debt, and Emotional Smells, we conducted a survey-based study of 72 software developers and analyzed the resulting data using Partial Least Squares Structural Equation Modeling (PLS-SEM). Our results show that burnout significantly contributes to the accumulation of Emotional Debt, which in turn positively influences the emergence of Emotional Smells. Moreover, Emotional Debt partially mediates the relationship between burnout and repository-level behavioral anomalies. These findings provide initial empirical evidence that developers' psychological well-being leaves measurable traces in software development activities and suggest new opportunities for non-intrusive monitoring approaches to improve both developer well-being and project sustainability.

Index Terms—TBD

I. INTRODUCTION

Unlike traditional industrial manufacturing, software development is characterized as a highly intensive cognitive activity [1]. Modern software engineering requires developers to operate in a continuous flow of both synchronous and asynchronous communication, demanding constant context switching and high mental agility [2]. While historical industrial labor measured effort through physical and muscular fatigue, the primary expenditure in computer science is quantified as cognitive load. In this environment, developers frequently push their working memory to its absolute limits under increasingly constrained release timelines and aggressive market demands [3]. When this intense cognitive strain is combined with unmanaged, prolonged occupational stress, it precipitates the onset of *burnout* [4] *i.e.*, a chronic psychological syndrome that cannot be dismissed as a transient state of exhaustion.

Historically, the software industry has fostered and celebrated the “*hero culture*” [5]. Within this paradigm, developers receive high esteem and institutional praise for working excessive hours, responding to incidents late at night, and sacrificing their weekends to meet deadlines. This glorification

of overwork is often normalized as a professional virtue. However, from organizational and psychological perspectives, this continuous state of hyper-activation is a primary catalyst for systemic burnout, undermining the fundamental needs of individuals as conceptualized in foundational psychological frameworks, such as Maslow’s hierarchy of needs [6].

Our work builds upon a core assumption derived from psychological and behavioral sciences, *i.e.*, individuals experiencing compromised mental well-being or chronic occupational distress inadvertently leave objective, observable traces of their psychological state within their behavioral outputs. Translating this paradigm into the socio-technical ecosystem of software development, we argue that a developer suffering from burnout leaves distinct, quantifiable traces within the source code, version control systems, and collaborative repositories. Consequently, it becomes possible to empirically map psychological distress to specific metric signatures captured directly from software development repositories.

To operationalize this phenomenon, this paper introduces and formalizes the concept of *Emotional Debt* as a counterpart to traditional technical debt, arguing that psychological friction incurs a long-term penalty on project sustainability. Specifically, the primary objectives of this study are fourfold:

- To formally define and conceptualize the construct of Emotional Debt within the software engineering domain;
- To identify, categorize, and operationalize its observable behavioral manifestations within software repositories, which we term *Emotional Smells*;
- To empirically investigate and validate the structural relationships linking developer burnout, Emotional Debt, and Emotional Smells through a rigorous structural equation modeling approach.
- A publicly available replication package that researchers can use to replicate and extend our work [?].

By bridging organizational psychology with empirical software engineering, this research aims to disassemble the hazardous myths surrounding the “*superhero culture*” in tech, providing project managers and development teams with early-warning, non-intrusive indicators to safeguard both developer well-being and software quality.

II. BACKGROUND AND RELATED WORK

This section reviews the theoretical foundations of occupational burnout and establishes the context of non-technical debt metaphors within the software engineering domain.

A. Occupational Burnout: Definitions and Economic Impact

The World Health Organization (WHO)¹ formally defines burnout as an occupational phenomenon resulting from chronic workplace stress that has not been successfully managed. Rather than being classified as a medical condition, it is conceptualized as a psychological syndrome structured around three distinct dimensions, as operationalized by the Maslach Burnout Inventory (MBI) [7]:

- *Emotional Exhaustion*: This dimension represents the core component of the syndrome, characterized by feelings of being emotionally overextended, drained, and psychologically depleted by professional demands. It reflects the direct toll of prolonged cognitive fatigue.
- *Cynicism and Depersonalization*: This dimension manifests as a defensive, detached, or callous response to work, colleagues, or clients. In software development, it often manifests as emotional detachment from the project, loss of engagement, and a cynical attitude toward collaborative practices.
- *Reduced Professional Efficacy*: This component entails a diminished sense of competence and accomplishment. The individual experiences a perceived decline in task performance, leading to feelings of inadequacy and an inability to deliver meaningful engineering solutions relative to historical personal baselines.

The broader societal and macroeconomic consequences of these unmanaged psychological strains are catastrophic. Global projections estimate that the collective cost of mental health conditions and their related operational consequences will exceed \$6 trillion USD by 2030 [8]. In high-tech industries, where cognitive assets are the primary driver of production, quantifying the downstream technical effects of this psychological erosion is paramount.

B. The Taxonomy of Debt in Software Engineering

The metaphor of “debt” has a rich history in software engineering, tracking how short-term shortcuts incur long-term compound penalties. Originally formalized as *Technical Debt* [9], the paradigm captured architectural flaws and poor code quality that hinder future velocity. Over the last two decades, empirical research has expanded this taxonomy beyond purely structural components [10], [11], [12], [13]. Researchers identified *Social and Community Debt* (often operationalized as community smells) [14], [15] **Euromicro**, which captures sub-optimal team structures, communication silos, and organizational friction that degrade collaboration. More recently, the community has introduced the concept of *Ethical Debt*, identifying the long-term societal, fairness,

and bias penalties incurred when architectural and algorithmic choices neglect ethical considerations [16], [17], [18].

Given that technical, social, and ethical debt systematically accumulate, distorting engineering decisions and degrading source code quality, it is reasonable to assume that psychological debt follows a parallel trajectory. While community debt captures the *structural* flaws of a team, it neglects the underlying *affective and psychological* state of its members. If a developer operates in chronic distress, their cognitive distortions will inevitably alter technical choices, introduce code anomalies, and compromise project longevity. However, a significant gap remains in the empirical literature:

Limitation

While technical and social debt have well-defined repository proxies (e.g., code smells, network silos), a comprehensive framework to map, accumulate, and operationalize psychological friction onto software artifacts has yet to be established.

III. RESEARCH METHOD

The *goal* of this study is to formalize and empirically validate the concept of Emotional Debt. The *purpose* is to characterize the mechanism by which chronic stress among developers (burnout) is related to the quality of software processes and artifacts. The *perspective* is for both researchers and practitioners interested in understanding the socio-technical nature of software development and in having a framework to more accurately and efficiently identify early warning signs of risk (Emotional Smells) within repositories, based on team members’ emotional and psychological states.

In accordance with our objective, we formulate four Research Questions (*RQs*):

Q RQ₁. *How can Emotional Debt be conceptualized in Software Engineering, and through which observable manifestations can it be identified?*

The primary objective of *RQ₁* is to formalize the construct of Emotional Debt within the software engineering domain and to identify its observable behavioral proxies, termed Emotional Smells.

Q RQ₂. *To what extent does developers’ perceived Burnout contribute to the accumulation of Emotional Debt?*

This stage of our research investigates the link between developers’ psychological state and the formation of socio-technical debt, examining how chronic occupational stress is a primary driver of debt accumulation.

Q RQ₃. *To what extent does Emotional Debt relate to the manifestation of Emotional Smells in the software development process?*

RQ₃ explores the downstream consequences of this accumulated psychological burden, analyzing its direct impact on technical artifacts and developer operational behaviors.

¹<https://www.who.int/>

Q RQ₄. Does Emotional Debt mediate the relationship between Burnout and Emotional Smells?

The final research question aims to clarify the structural mechanism by which psychological distress affects technical outcomes and to verify whether Emotional Debt acts as a critical intervening variable.

The research method is visually summarized in Figure X.

IV. RQ1 – CONCEPTUALIZATION OF THE EMOTIONAL DEBT

To ensure conceptual consistency with existing literature on Technical and Social Debt, we model the lifecycle of Emotional Debt through four sequential stages, *i.e.*, (1) *Decision Phase*, triggered by chronic stress and Burnout; (2) *Accumulation*, where unresolved emotional tension persists as a latent psychological burden; (3) *Observable Signals*, where Emotional Smells act as repository proxies; and (4) *Long-term Costs*, representing the structural “interest” paid in degraded maintainability. In the proposed model, the inception of the debt lifecycle is driven by developer burnout. Rather than viewing burnout as a static, individual psychological state, this framework conceptualizes it using one of the most established definitions provided from the psychological literature that defines burnout as a progressive socio-technical factor that evolves through three stages *i.e.*, *idealistic enthusiasm*, *stagnation*, *frustration*, and *apathy* [19], [20], [20].

These shifts directly interfere with a developer’s cognitive bandwidth, altering levels of attention, motivation, and task engagement. Consequently, burnout acts as a systemic catalyst that biases technical and organizational decision-making, leading to the accumulation of technical debt. As developers transition from initial engagement to emotional exhaustion (idealistic enthusiasm), they may over-engineer solutions that become difficult to maintain later. However, as they hit a wall and experience depersonalization (stagnation and frustration), their growing resentment leads to suboptimal architectural choices and reduced diligence during code development. Finally, as they slip into diminished professional efficacy (apathy), a detached attitude prompts superficial peer reviews and a general breakdown of rigorous engineering practices.

Crucially, this accumulation does not immediately manifest as explicit software defects; instead, it generates a latent socio-technical vulnerability that progressively undermines the integrity of both the software artifacts and team interactions. Formally, we can model the cumulative Emotional Debt D_E at a given time T as a function of the accumulated psychological stress and sub-optimal decisions over time:

$$D_E(T) = \int_0^T [f(\text{Exhaustion}) + g(\text{Cynicism}) + h(\text{Inefficacy})] dt \quad (1)$$

Consistent with technical debt metaphors, the final phase involves accruing interest and long-term costs. While the empirical validation of these costs lies outside the immediate scope of this study, theoretical modeling suggests they

manifest as a degradation of overall product quality, increased technical complexity, erosion of team cohesion, and eventual loss of project sustainability.

A. Operationalization via Emotional Smells

The core of this study lies in the manifestation phase, in which the latent Emotional Debt becomes traceable via Emotional Smells (S). These smells are defined as observable proxies in software repositories that signal the underlying debt accumulation. We formalize the total visibility of Emotional Debt as a set of multidimensional indicators mapped directly from the three dimensions of burnout defined by the MBI:

$$S = \{S_{\text{Exhaustion}}, S_{\text{Cynicism}}, S_{\text{Inefficacy}}\} \quad (2)$$

To operationalize these dimensions within Mining Software Repositories (MSR) contexts without relying on fragmented taxonomies, we categorize the metrics into three dense, thematic macro-dimensions:

Emotional Exhaustion Indicators ($S_{\text{Exhaustion}}$) quantify the alteration of work rhythms and cognitive overload. This macro-dimension is captured through *Persistent Off-Hours Activity* (recurrent commit patterns executed outside standard working hours, signaling workload overflow), *Workload Compression* (a significant reduction in the average size of commits, indicating task fragmentation), and *High-Frequency Commit Patterns* (rapid bursts of micro-commits within brief intervals where $\Delta t \rightarrow 0$, associated with acute time pressure).

Cynicism and Depersonalization Indicators (S_{Cynicism}) reflect socio-technical detachment and a decline in communicative care within the repository. This dimension is measured via *Negative Commit Sentiment* (the proliferation of pessimistic or hostile language in commit logs), *Self-Referential Commits* (vague or self-centered commit descriptions), *Communication Withdrawal* (a quantifiable drop in code comments, pull request reviews, and collaborative discussions), and *Lone Contribution Patterns* (isolated development trajectories restricted to specific modules without peer collaboration).

Reduced Professional Efficacy Indicators ($S_{\text{Inefficacy}}$) directly evaluate the drop in technical quality and engineering stability. It is operationalized through *Repeated Rework* (frequent alterations made to the exact same code segments within short temporal windows), *Repeated Reverts* (anomalously high frequencies of code rollbacks), *Non-Resolving Fixes* (successions of incremental patches that fail to definitively resolve the underlying defect), and *Uncritical AI Code Adoption* (over-reliance on generative AI coding assistants characterized by minimal or absent human verification, reflecting cognitive apathy).

B. Statistical Evaluation of the PLS-SEM Model

The theoretical robustness of RQ_1 is established through the evaluation of the measurement model, which assesses whether Emotional Smells reliably and validly represent the latent construct of Emotional Debt. Following the guidelines proposed by Hair *et al.* [21] the empirical evaluation is conducted

through a structured two-stage approach that assesses both the measurement and structural models.

Evaluation of the Measurement Model. The measurement model is first examined to verify the quality of the construct operationalization. This evaluation includes: (i) indicator reliability, assessed through outer loadings that should exceed 0.70; (ii) internal consistency reliability, evaluated using the Composite Reliability (ρ_C) and rho_A [22], with recommended values above 0.70; (iii) convergent validity, assessed through the Average Variance Extracted (AVE) [23], which should be greater than 0.50, ensuring that Emotional Smells effectively converge on the Emotional Debt construct; and (iv) discriminant validity, evaluated using the Heterotrait-Monotrait (HTMT) ratio [24], typically expected to remain below 0.85–0.90. The HTMT analysis is particularly important for demonstrating that Emotional Debt is empirically distinct from Burnout, thereby supporting its interpretation as an independent socio-technical construct rather than a mere psychological proxy.

Evaluation of the Structural Model. After validating the measurement model, the structural model is assessed to examine the hypothesized relationships among constructs. Following Hair *et al.* [21], this phase includes: (i) the analysis of multicollinearity among predictor constructs through the Variance Inflation Factor (VIF) [25], with values below 5 considered acceptable; (ii) the estimation of path coefficients [26]; (iii) the assessment of their statistical significance through bootstrapping procedures [26]; (iv) the evaluation of the model's explanatory power using R^2 [26]; and (v) the analysis of indirect effects and potential mediation relationships [26].

C. Empirical Analysis of the PLS-SEM Model

To empirically test our conceptual model, we conducted a survey-based study in accordance with the guidelines for software engineering research proposed by Kitchenham *et al.* [27]. The process was structured into four distinct phases: (i) *Survey Instrument and Construct Operationalization*, (ii) *Pilot Study*, (iii) *Participant Recruitment and Sampling*, and (iv) *Data Preparation and Cleaning*.

1) *Survey Instrument and Construct Operationalization:* The research instrument was a structured questionnaire developed on the Qualtrics platform,² consisting of 49 items organized into seven functional sections designed to map individual psychological states to specific technical behaviors.

The *Burnout* construct was operationalized using three clinical dimensions discussed previously, *i.e.*, *emotional exhaustion*, *cynicism*, and *professional inefficacy*. We evaluated each dimension using a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree).

The *Emotional Debt* was operationalized as 6 items developed *ex novo* based on the metaphor of *debt* in Software Engineering. These items capture the accumulation mechanism (tension that persists beyond task completion) and the awareness of operational deterioration (frustration influencing decision-making), evaluated on a 7-point Likert scale.

Emotional Smells. The most innovative part involves mapping these dimensions to observable technical behaviors in *Emotional Smells*. Process Smells map to Exhaustion (workload compression and off-hours activity); Communication Smells map to Cynicism (socio-technical withdrawal and negative sentiment); Technical Shortcut Smells map to Inefficacy (repeated rework, non-resolving fixes, and uncritical AI adoption). These behavioral patterns were measured using a 5-point frequency Likert scale (1 = Never, 5 = Always). The item mapping and descriptions are summarized in Table I. To ensure data integrity, attention checks [28] were included in our survey. Item order within each construct was fully randomized to mitigate response bias [29]. Furthermore, automated termination logic excluded respondents who were not active developers or lacked recent version control experience.

2) *Pilot Study:* Prior to large-scale distribution, a pilot study was conducted with ten industry professionals recruited through our personal network. The primary objectives were to verify the clarity and comprehensibility of the technical and psychological terminology, assess the internal consistency of the items' phrasing, and measure the average completion time to prevent respondent fatigue. The pilot phase resulted in minor lexical refinements to improve the precision of the questions and confirmed the feasibility of the survey duration.

3) *Participant Recruitment and Sampling:* The target population consisted of software developers. Recruitment was performed through three main channels, *i.e.*, via LinkedIn,³ internal networks, and using the Prolific platform.⁴ To ensure a high-quality, diverse sample, strict eligibility filters were applied on Prolific, restricting access to individuals employed in the Information Technology sector with specific engineering roles (*i.e.*, Software Engineer, Developer) and maintaining a historical platform approval rate between 95% and 100%.

4) *Data Preparation and Cleaning:* To ensure the integrity of the final dataset, a data cleaning procedure was implemented. The following exclusion criteria were applied:

- **Incompleteness:** We removed entries that did not reach the end of the questionnaire.
- **Speed-pacing:** We excluded responses completed in a duration significantly lower than the baseline since they could indicate low engagement.
- **Reliability Filters:** We removed participants who failed the attention checks or exhibited straight-lining response patterns.
- **Measurement Model Refinement:** During the empirical evaluation of the measurement model, indicator **QID5_7** (*Lone Contribution Pattern*) was removed from the *Emotional Smells* construct. The exclusion was necessitated by an insufficient outer loading (< 0.50), which severely undermined the convergent validity and statistical integrity of the construct, reducing the Average Variance Extracted (AVE) below acceptable standards.

At the end of this step, we obtained 72 answers.

²<https://www.qualtrics.com/>

³<https://www.linkedin.com/>

⁴<https://www.prolific.com/>

TABLE I
SURVEY ITEMS AND CONSTRUCT MAPPING

Construct / Dimension	QID	Item Description
<i>Burnout – Exhaustion</i>	QID1_1	I feel mentally exhausted because of my software development work.
	QID1_2	I feel emotionally drained after working.
	QID1_3	I feel exhausted at the end of the working day.
<i>Burnout – Cynicism</i>	QID2_1	I feel more detached from my work than in the past.
	QID2_2	I feel less emotionally involved in my project.
	QID2_3	I tend to distance myself from work-related interactions.
<i>Burnout – Inefficacy</i>	QID3_1	I feel less effective in my development activities.
	QID3_2	Tasks that were simple before now require much more effort.
	QID3_3	I doubt the quality of my contributions.
Emotional Debt	QID4_1	I carry unresolved work-related emotional tensions over time.
	QID4_2	Emotional stress from previous jobs continues to affect me.
	QID4_3	I work without addressing the emotional difficulties related to work.
	QID4_4	Accumulated frustration influences my work decision-making.
	QID4_5	Emotional tension persists even after tasks are completed.
	QID4_6	I feel emotionally “in debt” due to prolonged work pressure.
Emotional Smells	QID5_1	I have repeatedly worked at night or during weekends.
	QID5_2	I made many small commits instead of few well-structured ones.
	QID5_3	I performed commits with high frequency in short intervals.
	QID5_4	My commit messages expressed frustration or negativity.
	QID5_5	I wrote commit messages with little or no explanation (e.g., “fix”).
	QID5_6	I avoided commenting, reviewing, or participating in discussions.
	QID5_7	I worked on the same files or modules predominantly alone.
	QID5_8	I reworked the same code multiple times in a short period.
	QID5_9	I repeatedly reverted recent changes.
	QID5_10	I applied fixes that did not fully resolve the problem.
	QID5_11	I used AI-generated code with minimal review or adaptation.

D. Hypotheses Formulation

To address RQs from RQ_2 to RQ_4 , we formalize the relationships through four hypotheses, as shown in Figure 1.

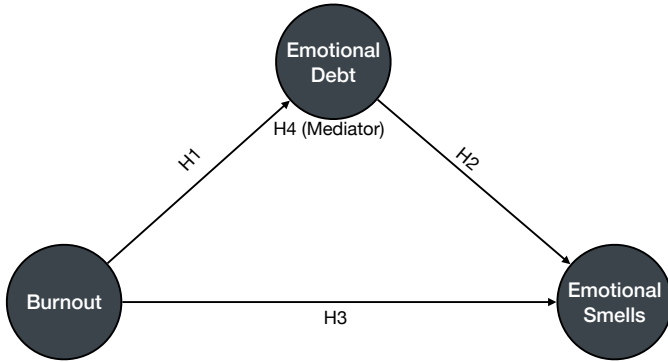


Fig. 1. Visual Conceptualization of our Hypotheses Formulation.

H1 (Burnout → Emotional Debt): The level of chronic burnout perceived by software developers has a positive and significant effect on the accumulation of Emotional Debt (RQ_2).

H2 (Emotional Debt → Emotional Smells): Accumulated Emotional Debt exerts a positive and significant effect on the manifestation of observable Emotional Smells within software development repositories (RQ_3).

H3 (Burnout → Emotional Smells): Burnout acts as a direct positive predictor of Emotional Smells, partially bypassing the latent structural debt mechanism (RQ_3).

H4 (Mediation Effect): Emotional Debt significantly mediates the relationship between developer Burnout and the behavioral manifestation of Emotional Smells (RQ_4).

V. H1: IMPACT OF BURNOUT ON EMOTIONAL DEBT ACCUMULATION

To evaluate hypothesis **H1**, which posits that developer burnout has a positive and significant effect on Emotional Debt accumulation, the relationship was analyzed using the PLS algorithm [26] to estimate the structural (inner) model. The core rationale suggests that the exhaustion and cynicism typical of burnout do not remain confined to the individual’s psychological sphere but translate into a latent debt that burdens the development process.

The analytical approach for RQ_2 began by estimating path coefficients (β) to assess both the strength and direction of the relationship between Burnout and Emotional Debt. The significance of these relationships was assessed using a non-parametric bootstrapping procedure with 5,000 subsamples, which provided empirical estimates of T-values and p-values. Before interpreting the structural results, collinearity was examined through the Variance Inflation Factor (VIF) to ensure that multicollinearity did not distort the model estimates. Finally, the model’s explanatory power was evaluated using

the Coefficient of Determination (R^2), which indicates how much variance in Emotional Debt is explained, together with Cohen's f^2 , which quantifies the specific effect size of burnout on the endogenous variable.

VI. H2: EMOTIONAL DEBT AND EMOTIONAL SMELLS

To test hypothesis **H2**, which states that Emotional Debt significantly predicts the occurrence of Emotional Smells, the linear path from debt to smells was evaluated within the PLS-SEM framework.

Beyond path significance (β) and effect size (f^2) analyses, a distinctive feature for answering RQ_3 was the assessment of the model's out-of-sample predictive power using the PLSpredict procedure in line with Shmueli *et al.* [30]. This execution involved a data-splitting strategy to systematically evaluate whether the structural model can accurately predict the "smells" indicators for observations not used during the model estimation phase.

The predictive relevance was validated by directly comparing the Root Mean Square Error (RMSE) [31] of the PLS model with that of a native linear regression benchmark (LM).

A. H3: Direct Impact of Burnout on Emotional Smells

H3 asserts that Burnout has a positive and significant direct effect on the occurrence of Emotional Smells. Theoretically, this path explores the "instantaneous" translation of *exhaustion*, *cynicism*, and *inefficacy* into observable technical anomalies, such as fragmented commits or negative sentiment, without necessarily requiring the buildup of a latent state of debt.

The evaluation of this relationship within the PLS-SEM framework involved estimating a direct linear path from the Burnout construct to Emotional Smells to measure the immediate impact of occupational stress on development behavior. As with other structural paths, a non-parametric bootstrapping procedure with 5,000 subsamples was applied to calculate the stability of the relationship and verify its statistical significance.

B. H4: The Mediating Role of Emotional Debt

The final part aims to clarify the structural mechanism through which psychological distress translates into technical outcomes, testing hypothesis **H4** (the mediation of Emotional Debt between Burnout and Smells). The analysis decomposes the effect into its constituent parts: the *Direct Effect* (the path from Burnout to Smells while controlling for the mediator) and the *Indirect Effect* (the product of the paths Burnout $\rightarrow$ Emotional Debt and Emotional Debt $\rightarrow$ Emotional Smells).

Significance testing was conducted using a non-parametric bootstrapping procedure with 5,000 subsamples to compute p-values without distributional assumptions.

VII. RESULTS

Before discussing the structural model, we outline the demographic and professional characteristics of our sample.

- **Gender & Geography:** The sample is male-majority, with 55 identifying as Male, 11 as Female, 1 as Non-Binary, and 5 who preferred not to respond. The geographical distribution spans multiple regions, with participation intensity ranging from 1 to 18 respondents per area.
- **Age Distribution:** The majority of participants fall within the core professional age bracket of 25–34 years (38 respondents, 52.8%), followed by the 35–44 range (22 respondents, 30.6%), while younger and older groups represent minor fractions.
- **Professional Expertise:** In contrast to the age profile, the professional experience is highly balanced and leans heavily toward seniority. The largest single segment consists of software engineers with more than 10 years of active industry experience (19 respondents, 28.4%), ensuring that the data reflects a workforce familiar with mature software evolution practices.

In accordance with the structural equation modeling guidelines proposed by Hair *et al.*, the empirical analysis is presented in two sequential stages: (i) the Evaluation of the Measurement Model, and (ii) the Evaluation of the Structural Model.

A. RQ1 – Evaluation of the Measurement Model

The measurement model establishes whether the operationalized indicators reliably and validly represent their respective latent constructs.

TABLE II
OUTER LOADINGS FOR BURNOUT, EMOTIONAL DEBT, AND EMOTIONAL SMELLS CONSTRUCTS.

Item	Burnout	Item	Emotional Debt	Item	EmotionalSmells
QID1_1	0.814	QID4_1	0.838	QID5_1	0.620
QID1_2	0.849	QID4_2	0.803	QID5_2	0.593
QID1_3	0.817	QID4_3	0.677	QID5_3	0.680
QID2_1	0.787	QID4_4	0.738	QID5_4	0.710
QID2_2	0.784	QID4_5	0.789	QID5_5	0.745
QID2_3	0.739	QID4_6	0.858	QID5_6	0.833
QID3_1	0.784			QID5_8	0.782
QID3_2	0.777			QID5_9	0.811
QID3_3	0.673			QID5_10	0.762
				QID5_11	0.680

As summarized in Table II, the outer loadings for the *Burnout* indicators range from 0.673 to 0.849, with the majority comfortably exceeding the recommended 0.70 threshold. Although indicator QID3_3 (0.673) falls slightly below this value, it was retained to preserve the multifaceted definition of reduced professional efficacy without compromising internal consistency.

For the *Emotional Debt* construct, loadings range from 0.677 to 0.858. The marginal loading of QID4_3 (0.677) was accepted due to the exploratory and novel nature of this construct in Software Engineering.

Regarding *Emotional Smells*, the loadings exhibit higher behavioral heterogeneity (ranging from 0.593 to 0.833). Indicators QID5_1 (0.620) and QID5_2 (0.593) present the

lowest values. Retaining these moderate items is conceptually justified to maintain the broad socio-technical scope of behavioral patterns traccabili nei repository, preventing excessive semantic reduction.

TABLE III
INTERNAL CONSISTENCY AND CONVERGENT VALIDITY (RQ1)

Construct	Cronbach's α	rho_A	Comp. Reliability (ρ_C)	AVE
Burnout	0.920	0.928	0.934	0.612
Emotional Debt	0.875	0.887	0.906	0.618
Emotional Smells	0.899	0.911	0.917	0.526

As shown in Table III, all latent constructs demonstrate robust internal consistency and convergent validity. *Emotional Debt* achieved a high Composite Reliability (ρ_C) of 0.906, a Cronbach's Alpha of 0.875, and an AVE of 0.618, confirming that the construct captures roughly 62% of the variance of its underlying metrics. Similarly, *Emotional Smells* demonstrates adequate convergence metrics ($\rho_C = 0.917$, AVE = 0.526).

TABLE IV
DISCRIMINANT VALIDITY (HTMT RATIO)

Relationship	HTMT	2.5% CI	97.5% CI
ED $\leftrightarrow$ Burnout	0.846	0.730	0.941
ES $\leftrightarrow$ Burnout	0.652	0.474	0.807
ES $\leftrightarrow$ ED	0.711	0.573	0.841

Discriminant validity was established via the Heterotrait-Monotrait (HTMT) ratio (Table IV). All values strictly remain below the conservative psychometric threshold of 0.85. Crucially, the 95% bootstrapping confidence interval for the Emotional Debt $\leftrightarrow$ Burnout link (0.730, 0.941) This statistically supports our conceptual framework, proving that Emotional Debt is an independent socio-technical construct rather than a redundant psychological proxy.

RQ₁ Summary

The results of the measurement model confirm the validity and reliability of the construct, supporting its conceptualization as an accumulation of unprocessed emotional tension that is reflected in behaviors observable during the developmental process.

B. Structural Model Evaluation

Following measurement validation, we evaluated the inner model to test the structural relationships. Collinearity was examined using the Variance Inflation Factor (VIF) to prevent structural distortion.

As shown in Table V, the VIF between Burnout and Emotional Debt is 1.000, since Burnout acts as the lone structural predictor of debt accumulation. For the paths predicting Emotional Smells, the VIF stands at 2.586 for both predictors. Because these values fall well below the conservative threshold of 3.3, it confirms the structural stability of the model, proving

TABLE V
COLLINEARITY ASSESSMENT (VIF RATIO)

Relationship	VIF (O)	2.5%	97.5%
Burnout $\rightarrow$ EmotionalDebt	1.000	1.000	1.000
Burnout $\rightarrow$ EmotionalSmells	2.586	1.934	4.146
EmotionalDebt $\rightarrow$ EmotionalSmells	2.586	1.934	4.146

that the predictors do not overlap to a degree that would destabilize the path coefficients.

TABLE VI
STRUCTURAL PATH COEFFICIENTS.

Relationship	Coefficient (β)	T-value	p-value
Burnout $\rightarrow$ EmotionalDebt	0.783	17.233	0.000
Burnout $\rightarrow$ EmotionalSmells	0.618	7.938	0.000
EmotionalDebt $\rightarrow$ EmotionalSmells	0.432	2.670	0.008

TABLE VII
COEFFICIENTS OF DETERMINATION (R^2) FOR ENDOGENOUS CONSTRUCTS

Endogenous Construct	R^2	T-value	p-value
EmotionalDebt	0.613	8.624	0.000
EmotionalSmells	0.454	5.713	0.000

1) *RQ2 – Contribution of Burnout to Emotional Debt Accumulation (H1)*: The inner model estimates provide strong empirical support for hypothesis **H1**. The path coefficient indicates a positive relationship between Burnout and Emotional Debt ($\beta = 0.783$, $T = 17.233$, $p < 0.001$, see Table VI).

The model demonstrates robust explanatory power for the latent accumulation process, *i.e.*, the coefficient of determination ($R^2 = 0.613$, Table VII) shows that chronic burnout alone accounts for 61.3% of the total variance in Emotional Debt. Furthermore, the effect size is classified as *extremely large* ($f^2 = 1.586$). This confirms that removing Burnout from our model would critically strip away its ability to explain debt formation, designating occupational distress as the fundamental catalyst of latent socio-technical burden.

RQ₂ Summary

Burnout has a positive and highly significant effect on Emotional Debt ($\beta = 0.783$, $p < 0.001$). The R^2 value of 0.613 indicates that burnout explains approximately 61.3% of the variance in emotional debt, highlighting a substantial impact. Furthermore, the effect size ($f^2 = 1.586$) confirms the large effect.

2) *RQ3 – Manifestation of Emotional Smells and its Relationship with Emotional Debt (H2)*: The evaluation of the structural model also supports hypothesis **H2**. As shown in Table VI, Emotional Debt exerts a positive and statistically significant effect on the occurrence of Emotional Smells ($\beta = 0.432$, $T = 2.670$, $p = 0.008$).

The independent effect size of debt on smells is small-to-moderate but significant ($f^2 = 0.132$). Combined with Burnout, the structural model successfully explains 45.4% of the overall variance in repository Emotional Smells ($R^2 = 0.454$, Table VII). This indicates that nearly half of the tracked technical anomalies and communication shortcuts can be traced directly back to the psychological state of the development team.

To establish practical operational utility, we ran an out-of-sample predictive assessment via the *PLSPredict* procedure.

To establish practical operational utility, we executed an out-of-sample predictive assessment via the *PLSPredict* procedure shown in Table VIII.

TABLE VIII
COMPLETE OUT-OF-SAMPLE PREDICTIVE POWER FOR SURVEY ITEM INDICATORS.

Indicator	$Q^2_{predict}$	PLS RMSE	LM RMSE
QID4_1	0.472	1.244	1.252
QID4_2	0.293	1.539	1.634
QID4_3	0.183	1.570	1.782
QID4_4	0.241	1.488	1.562
QID4_5	0.413	1.384	1.509
QID4_6	0.487	1.287	1.148
QID5_1	0.147	1.254	1.365
QID5_10	0.204	1.060	1.100
QID5_11	0.088	1.354	1.485
QID5_2	0.129	0.939	1.056
QID5_3	0.011	1.004	1.067
QID5_4	0.034	1.227	1.338
QID5_5	0.182	1.241	1.403
QID5_6	0.365	0.949	0.983
QID5_8	0.297	0.894	0.959
QID5_9	0.237	0.970	1.015

As summarized in Table VIII, all $Q^2_{predict}$ values across the entire behavioral indicator block are strictly positive, confirming out-of-sample predictive capability. Furthermore, the RMSE outperforms the naive linear regression benchmark across almost all tracked behavioral metrics (*e.g.*, QID5_6, QID5_8, QID5_9), confirming its role as an empirical early-warning diagnostic tool in engineering repositories.

RQ₃ Summary

Emotional Debt has a positive and significant effect on Emotional Smells ($\beta = 0.432$, $p = 0.008$). In terms of contribution to explained variance, the model shows that Burnout and Emotional Debt together explain 45.4% of the variance in Emotional Smells ($R^2 = 0.454$). The effect size of Emotional Debt ($f^2 = 0.132$) indicates a low-to-moderate but significant contribution.

C. RQ4 – The Intervening Mechanism of Emotional Debt (H3 & H4)

To uncover the exact socio-technical mechanism linking psychological strain to technical degradation, we conducted a bootstrapping procedure (5,000 subsamples) to assess the

direct impact of Burnout (**H3**) versus the mediated path via Emotional Debt (**H4**).

TABLE IX
SUMMARY OF MEDIATION ANALYSIS.

Relationship	Path Coefficient (β)	T-value	p-value	Result
Total Effect	0.956	–	–	–
Direct Effect (H_3)	0.618	7.938	0.000	Significant
Indirect Effect (H_4)	0.338	2.614	0.009	Significant
Mediation Type: Partial Mediation				

As shown in Table IX, the analysis shows a *partial mediation*. The structural *Indirect Effect* (Burnout $\rightarrow$ Emotional Debt $\rightarrow$ Emotional Smells) is positive and statistically significant ($\beta = 0.338$, $T = 2.614$, $p = 0.009$), validating **H4**. This suggests that a massive share of developer burnout damages repositories slowly through the progressive buildup and storage of lingering, unaddressed emotional load.

However, the *Direct Effect* of Burnout on Emotional Smells (**H3**) remains strong and highly significant ($\beta = 0.618$, $T = 7.938$, $p < 0.001$) even when controlling for the mediator.

RQ₄ Summary

The results indicate partial mediation. The indirect effect of burnout on emotional smells via emotional debt is positive and significant, while the direct path remains significant. This suggests that burnout is related to emotional smells through the progressive accumulation of emotional debt.

VIII. IMPLICATIONS

IX. THREATS TO VALIDITY

This section evaluates the primary threats to the validity of our study.

Internal Validity. The main limitation arises from the study’s cross-sectional design, as data were collected at a single point in time, which allows for the identification of statistical relationships but does not permit the definitive establishment of temporal causality. Additionally, reverse causality cannot be entirely ruled out, as it is theoretically possible that dysfunctional development behaviors or negative organizational conditions contribute to increased developer burnout rather than being its sole consequence. Unmeasured confounding variables, such as specific development methodologies or the use of generative AI tools, might also influence the model’s variables simultaneously.

Construct Validity. Because Emotional Debt is an emerging construct in Software Engineering, no established measurement scales exist, and some indicators may not capture all latent dimensions of the phenomenon. To mitigate this risk, the questionnaire was designed in accordance with the guidelines proposed by Kitchenham and Pfleeger [27], and the quality of the measurement model was empirically assessed using Average Variance Extracted (AVE) and the Heterotrait-Monotrait

(HTMT) ratio to ensure that Emotional Debt remains distinct from burnout.

External Validity. This threat concerns the generalizability of the findings. As the sample was recruited primarily through the Prolific platform, it may not fully represent the global population of software developers. Furthermore, the analysis relies on self-reported data based on subjective perceptions; future studies that incorporate longitudinal data and objective repository mining will be necessary to confirm the robustness of these results across diverse operational contexts.

Conclusion Validity. This threat concerns the reliability of the statistical inferences drawn from the data. Specifically, regarding the sample size of 72 valid responses, which might be insufficient to detect very small effects. This threat was addressed by employing the Partial Least Squares (PLS-SEM) technique, which is particularly robust for moderate sample sizes in exploratory research, and by using a nonparametric bootstrapping procedure with 5,000 subsamples to assess the stability and significance of all structural paths.

X. CONCLUSIONS

In this paper, we introduced the concept of Emotional Debt and investigated its relationship with developer burnout and repository-level behavioral indicators, namely Emotional Smells. Our results show that burnout significantly contributes to the accumulation of Emotional Debt, which, in turn, is associated with observable socio-technical anomalies. Furthermore, Emotional Debt partially mediates the relationship between burnout and Emotional Smells, providing evidence of a structural mechanism linking psychological distress to software engineering outcomes. As part of our future agenda, we want to investigate the automatic detection of Emotional Smells from software repositories and validate our findings through large-scale longitudinal studies.

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